

Gangadhar Renka

SR . Healthcare Data Analyst

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PROFESSIONAL SUMMARY

- Over 9 years of experience in **healthcare data analytics** spanning **payer, provider, pharmaceutical supply chain**, and **clinical research** domains, supporting **Medicare Advantage, Commercial**, and **Medicaid** populations
- Extensive hands-on experience in **regulatory and quality reporting**, including **CMS Star Ratings, HEDIS, MIPS**, and **pharma compliance**, covering 15M+ member records with 100% submission accuracy and measurable performance gains
- Proficient in **risk adjustment modelling** and **HCC coding**, developing **NLP-based optimization workflows** in **Python** and **SAS**, increasing RAF accuracy by 34% and enabling \$2.8M in additional reimbursement.
- Designed and deployed **predictive models** for **readmission risk, sepsis detection, cost forecasting**, and **demand planning**, leveraging **Random Forest, XG Boost**, and **logistic regression** with 85–94% accuracy and \$4.8M+ cost savings.
- Applied advanced **statistical methods** including **Cox regression, mixed-effects models**, and **propensity score matching (PSM)** to analyze **clinical outcomes, drug effectiveness**, and **treatment pathways** across 100K+ patients.
- Conducted **real-world evidence (RWE)** and **longitudinal cohort studies** using **R, SAS**, and **Python**, supporting 12+ peer-reviewed publications and contributing to 3 FDA submissions.
Designed and optimized **pharmaceutical supply chain analytics** across 18,000+ pharmacies, reducing stockouts by 28% and improving fill rates to 96.5% through inventory forecasting and distribution modelling
- Developed **executive dashboards** and **KPI scorecards** using **Power BI, Tableau**, and **R Shiny**, with automated alerting and role-based access across quality, pharmacy, finance, and care management functions.
- Architected **data integration pipelines** handling 50M+ records/month from **Epic Clarity, Cerner, claims, pharmacy**, and **SDOH** systems, ensuring compliance with **FHIR, HL7, HIPAA**, and **interoperability** standards via **Azure Data Factory, Databricks**, and **Snowflake**.
- Collaborated with **clinicians, underwriters, actuaries, pharmacists**, and **product managers** to translate business needs into analytics solutions, and led team training, peer reviews, and QA for scalable projects.
- Presented at industry conferences such as **HIMSS, AHIMA**, and **HFMA**, reviewed publications for top-tier journals, and mentored junior analysts on **analytics governance, protocol design**, and **research methodology**

TECHNICAL SKILLS:

Programming	SQL, Python, R, SAS, SPSS, MATLAB, Stata, Advanced Excel (with VBA Automation)
Python Libraries	Pandas, NumPy, Scikit-learn, TensorFlow, Kera's, NLTK, Matplotlib, Seaborn, Plotly
R Libraries	ggplot2, dplyr, survival, caret, shiny, tidy verse, lubricate
SAS Modules	Base, STAT, MACRO, Enterprise Guide (EG), DI Studio, Visual Analytics, Enterprise Miner, Forecast Studio
Healthcare Systems & EHRs	Epic (Clarity, Caboodle, Slicer Dicer, Chronicles, Healthy Planet), Cerner (PowerChart, Health Facts, Discern Analytics), Meditech (Magic, Expanse), Athena Collector, NextGen, All Scripts, eClinicalWorks
Pharmaceutical Systems	Cardinal Health Connect Rx, McKesson Relay Health, Pharmacy Management Systems
Healthcare Standards & Interoperability	HL7 FHIR R4, DICOM, X12 EDI, CCD/CDA Documents, Direct Trust Messaging, Health Information Exchange (HIE), Clinical Data Repository (CDR)
Clinical Coding & Standards	ICD-10-CM/PCS, CPT/HCPCS Level II, DRG/MS-DRG, APR-DRG, SNOMED CT, LOINC, Rx Norm, NDC Codes, DEA Numbers, AWP Pricing

Healthcare Quality & Compliance	HCC Risk Adjustment (V24, V28), RAF Score Calculation, CMS Quality Measures (CQM), HEDIS, Medicare Advantage Star Ratings, MIPS, CDS Hooks, Pharmaceutical Rebate Analytics
Statistical Methods & ML	Survival Analysis, Kaplan-Meier, Cox Models, Logistic/Linear Regression, GLMs, Mixed-Effects Modeling, Time Series (ARIMA), Propensity Score Matching, Random Forest, XG Boost, Neural Networks, SVM, K-means, NLP, Bayesian Analysis
Data Visualization & BI Tools	Power BI (DAX, Power Query, Custom Visuals), Tableau (LOD, Advanced Calculations, Server), SAS Visual Analytics, R Shiny, Qlik Sense
Database Management Systems	SQL Server (T-SQL, SSIS, SSAS, SSRS), Oracle (PL/SQL), MySQL, PostgreSQL, IBM Db2
Cloud & Big Data Tools	Snowflake, Azure SQL/Data Warehouse, Amazon Redshift, Google Big Query, Azure Health Data Services, AWS Health Lake, Google Healthcare API, Databricks, Apache Spark, Azure Data Factory, AWS Glue
Healthcare Analytics Domains	Claims & Encounter Data, Population Health, Clinical Outcomes, HEOR, Pharmacoeconomics, Revenue Cycle, Risk Adjustment, SDOH, Provider Analytics, Prior Authorization, PBM, DUR, Pharma Supply Chain, Inventory, Rebates

PROFESSIONAL EXPERIENCE:

Client: CVS Health (Client) -Austin, TX
Role: Sr. Health Care Data Analyst

January 2024– Present

Responsibilities:

- Integrated **claims, clinical, pharmacy, dental**, and **SDOH data** for **3.4M+ members** across **Medicare Advantage, Commercial, and Medicaid** using **Python** and **SAS**.
- Built scalable **data infrastructure** to process **medical, pharmacy, dental, and behavioral health data**, reducing latency from **45 days to 3 days** with **99.2% accuracy**.
- Developed **HCC risk adjustment models** using **NLP**, improving **coding accuracy** by **34%** and recovering **\$2.8M** in **risk-based revenue**.
- Designed and deployed **risk stratification models** with **Johns Hopkins ACG** and **machine learning algorithms**, achieving **88% accuracy** and saving **\$4.1M**.
- Applied **Cox proportional hazards models** to compare **treatment outcomes** across **oncology, rheumatology, and neurology** in **45,000+ members**.
- Conducted **cohort studies** in **450+ provider networks** using **survival analysis** and **propensity score matching**.
- Created **forecasting models** using **economic indicators** and **ML techniques**, reaching **94% accuracy** in **cost prediction**.
- Developed **cost-sharing impact models** to analyze **utilization patterns** and **financial exposure**.
- Modeled **cost-effectiveness** and **provider network efficiency** to support **value-based contracts** in **15+ markets**.
- Created **total cost of care models** for **finance** and **population health strategy** teams.
- Built **fraud detection models** using **Random Forest, XG Boost, and Neural Networks**, identifying **\$3.2M** in **fraudulent claims** and reducing **false positives** by **45%**.
- Built **predictive models** for **high-risk members**, reducing **ED visits** by **31%** and raising **medication adherence** to **84%**.
- Automated **CMS reporting (Star Ratings, HEDIS, ACA)**, reducing manual effort by **75%** and ensuring **100% compliance**.
- Built **HEDIS engine modules** to support **accreditation** and **incentive payments**.
- Designed **Star Ratings optimization models**, improving **plan scores** by **0.5 stars**.
- Created **dashboards** with **KPI tracking, role-based views, and automated alerts** for **leadership**.
- Implemented **data governance framework**, improving **data quality** from **87% to 98%**.
- Evaluated **provider network adequacy** to support **CMS compliance**.
- Performed **actuarial analysis** on **MLR, trend projections, and reserve estimates**.
- Built **NLP pipelines** for **provider notes** and **prior authorization forms**, accelerating **processing** by **23%**.

Technical Environment:

Python (Pandas, Scikit-learn, XG Boost, TensorFlow, NLTK), SAS (Base, EG, DI Studio), R (survival, tidy verse), SQL, Py Spark, Snowflake, Azure Data Factory, Databricks, SQL Server, Tableau, Power BI, HL7, FHIR, ICD-10, CPT, HEDIS, HCC, ACG, CMS Star Ratings, NLP, Cox Models, Propensity Score Matching, Survival Analysis, Data Governance, KPI Dashboards, Agile, Git, JIRA, Confluence

Client: Cleveland Clinic (Client) - Cleveland, OH**August 2022– Dec 2023****Role: Clinical Data Analyst****Responsibilities:**

- Conducted **biostatistical analysis** for **35+ clinical studies** and **NIH-funded research** across **cardiovascular, oncology, neurology, and organ transplantation** using **Epic Clarity** with **12M+ patient records**.
- Implemented **machine learning algorithms** for **early disease detection** using **ensemble methods** and **feature engineering** on **structured/unstructured EHR data**, achieving **90% sensitivity** and **85% specificity** for **high-risk cardiac patients**.
- Performed **survival analysis** and **longitudinal outcomes research** on **125,000+ patients** with **5–15-year follow-ups**, addressing **competing risks, time-varying exposures, and selection bias** via **advanced statistical modelling**.
- Designed **statistical analysis plans (SAPs)** and conducted **interim safety analyses** for **Phase II/III clinical trials**, complying with **FDA regulations** and **ICH-GCP guidelines**, collaborating with **Data Safety Monitoring Boards (DSMB)**.
- Built **real-world evidence (RWE) platforms** integrating **claims, EHR, and patient-reported outcomes** for **comparative effectiveness, post-market surveillance, and regulatory submissions** (pharma/device).
- Architected **clinical decision support systems (CDSS)** integrated with **Epic EHR workflows**, processing **2M+ encounters/year**, with **85%+ provider adoption**.
- Developed **NLP algorithms** for **clinical notes, radiology reports, and pathology findings** to support **automated risk scoring, quality reporting, and research data extraction**.
- Built **predictive models** for **sepsis detection, AKI prediction, and medication safety** using **real-time streaming data** from **monitors, labs, and pharmacy systems**.
- Implemented **computer vision** for **medical imaging analysis**, improving **diagnostic accuracy** across **CT, MRI, and X-ray**.
- Created **sepsis early warning algorithms** using **lab values, vital signs, and ML ensembles**, contributing to **18% mortality reduction, 2.3-day shorter LOS, and \$2.1M savings**.
- Developed **automated quality reporting infrastructure** supporting **CMS and Joint Commission metrics** with **99.5% audit-validated accuracy**.
- Built **readmission risk models** using **clinical, demographic, and SDOH factors**, achieving **c-statistic of 0.78**, reducing **readmissions by 18%**.
- Performed **physician performance and practice variation analysis** across **3,500+ medical staff** and **180+ departments**.
- Analyzed **HCAHPS and patient-reported outcomes** using **mixed-effects modeling** to identify **drivers of satisfaction and care quality**.
- Designed **clinical trial SAPs** and collaborated on **Phase II/III trials** under **FDA/DSMB guidance**.
- Implemented **research data management systems**, reducing **query rates by 60%** and accelerating **lock timelines by 3 weeks** for **15+ studies**.
- Created **interactive analytics platforms** for **real-time trial monitoring**, improving **completion timelines by 40%**.

Technical Environment:

Python (Scikit-learn, Pandas, TensorFlow), R (survival, lme4), SAS, SQL, Epic Clarity, NLP, EHR, CMS, HCAHPS, FDA, ICH-GCP, Cox Models, Ensemble Methods, Propensity Score Matching, Mixed-Effects Modelling, Computer Vision, Tableau, Power BI, Real-World Evidence (RWE), CDSS, SDOH, JIRA, Git, Clinical Training

- **Engineered robust backend modules** with **Java, Spring MVC, and Hibernate**, powering critical retail systems

Client: Cardinal Health - Waukegan, IL**June 2020 – July 2022****Role: Senior Healthcare Data Analyst**

- Analyzed supply chain data across **18,000+ pharmacy locations** and **2.4B+ prescription transactions** using **Python, SAS, and Snowflake**

- Created **demand forecasting models** using **time series & ML (92% accuracy)** → reduced stockouts by **28%**, improved fill rates to **96.5%**
- Developed **inventory optimization models** (seasonality + constraints) → reduced carrying costs by **22%**, maintained **98% service levels**
- Built **executive dashboards** for real-time KPI tracking and supply chain alerts
- Built **drug utilization & substitution algorithms** using **NLP** across **450+ health systems** → generated **\$3.7M** in cost savings
- Created **clinical decision support algorithms** for drug interactions and dosing → supported **8.5M+ patients**
- Performed **pharmaceutical outcomes research** using **survival analysis** for **2.1M+ patients** (oncology, biosimilars, specialty drugs)
- Processed **\$1.2B+** in manufacturer rebates via automated reconciliation
- Monitored **340B compliance** across **1,800+ covered entities** → maintained audit readiness and **\$180M annual savings**
- Automated **DEA, FDA, state reporting** → achieved **100% compliance**, reduced manual effort by **65%**
- I identified **\$2.8M in fraud** using ensemble/anomaly detection → reduced false positives by
- Evaluated **formulary positioning & drug pricing strategies**
- Modeled **cost-effectiveness** for specialty therapies and biosimilars adoption
- Implemented **data governance**, data quality, and master data management protocols

Client: McKesson Corporation -Irving, TX

Oct 2018– May 2020

Role: Healthcare Data Analyst

Responsibilities:

- Analysed distribution data for **22,000+ pharmacies**, processing **\$180B+ revenue**
- Developed **predictive models for shortage management** → 94% accuracy, reduced impact by **35%**
- Built **integrated dashboards** for ops, clinical programs, business development
- Created analytics platform for **oncology, immunology, rare diseases** → served **580K+ patients**
- Built **patient adherence models** (claims + clinical) → 89% sensitivity, improved adherence by **31%**
- Performed **RWE studies** across **45+ therapeutic areas** → supported **FDA submission**
- Executed **MTM & intervention analytics** → generated **\$4.2M in savings**
- Built **prior auth models** using **NLP + decision trees** → cut time by **45%**, approvals to **92%**
- Analysed **biosimilar adoption** across 12 categories → supported **\$25M revenue**
- Created **market intelligence platform** (pricing, competitors, therapeutic pipeline) for **500+ products**
- Evaluated **value-based contracts** using health economics modelling
- Automated **CMS Part D, Medicare compliance**, and quality metrics across **15,000+ locations**
- Applied data mining to identify therapeutic opportunities, market gaps, and customer expansion strategies

Technical Environment:

Python (NumPy, Pandas, Scikit-learn) | SAS (Enterprise Miner, Forecast Studio) | R (survival, caret) | SQL Server, Oracle, Databricks, AWS Redshift | Power BI, Tableau Server | Relay Health, AWP Pricing, MTM, Biosimilar Analytics, CMS Part D, NDC Mapping

Apollo Hospitals (Client)- Hyderabad-India

July 2016 – July 2018

Role: Healthcare Data Analyst

- Processed **EHR data** from **200+ hospitals** across **India, Bangladesh, Sri Lanka, Kuwait, Qatar, and Oman**, including records from **urban, rural, and medical tourism facilities**.
- Built **risk prediction models** for **chronic disease patients** using **logistic regression, decision trees, and ensemble methods**, reaching **87% accuracy** and lowering **inpatient stays** by **14%**.
- Evaluated **treatment cost-effectiveness** (e.g., **cardiac surgery vs. drug therapy**), reducing **cost variation** by **19%** across branches through model-driven standardization.
- Designed **automated dashboards** for **JCI readiness**, capturing **infection rates, adverse events, and satisfaction metrics** to support **regulatory audits**.
- Led **data migration** between **HMS and HIMS**, implemented **standard mapping protocols**, and created **integrity checks**, increasing **data completeness** by **45%** and **accuracy** by **38%**.
- Applied **queuing models** to simulate **patient movement** through **radiology, labs, and pharmacy**, cutting **wait times** from **90 to 35 minutes** and increasing **throughput** by **28%**.

- Wrote **advanced SQL queries** to pull **clinical, financial, and operational data**; automated reports using **VBA macros**, reducing **manual work by 50%**.
- Developed **Tableau dashboards** to track **hospital KPIs**, supporting real-time decisions in **staffing, quality, and clinical operations**.

Technical Environment:

SQL, Excel VBA, Tableau, Logistic Regression, Decision Trees, Ensemble Models, Queuing Models, HMS/HIMS, Data Mapping, JCI Standards, KPI Dashboards, Infection Metrics, Cost Analysis, Process Optimization